

KochiDocs

- Problem Statement ID – 25080
- Problem Statement Title - Document Overload at Kochi Metro Rail Limited (KMRL) - An automated solution
- Theme - Smart Automation
- PS Category- Software
- Team ID - TBD
- Team Name - Platform 0



KochiDocs : Centralized AI platform for KMRL. Automates document ingestion, OCR, organization & analysis. Eliminates manual searches, digitizes paper, and combats document overload.



Key Features



Automated Ingestion & OCR: Scans physical records using OCR (Tesseract/Google Vision) for digital text capture – enabling KMRL to go truly paperless.



Smart Classification & Tagging: Leverages NLP (spaCy, transformers) to categorize documents by type, department, date, etc., and extract key metadata (e.g. titles, dates).



Instant Search & Q&A: Indexes all content for lightning-fast retrieval. Staff can query in natural language (English/Malayalam) and get direct answers or summary of relevant files (like an internal “Google” for KMRL).

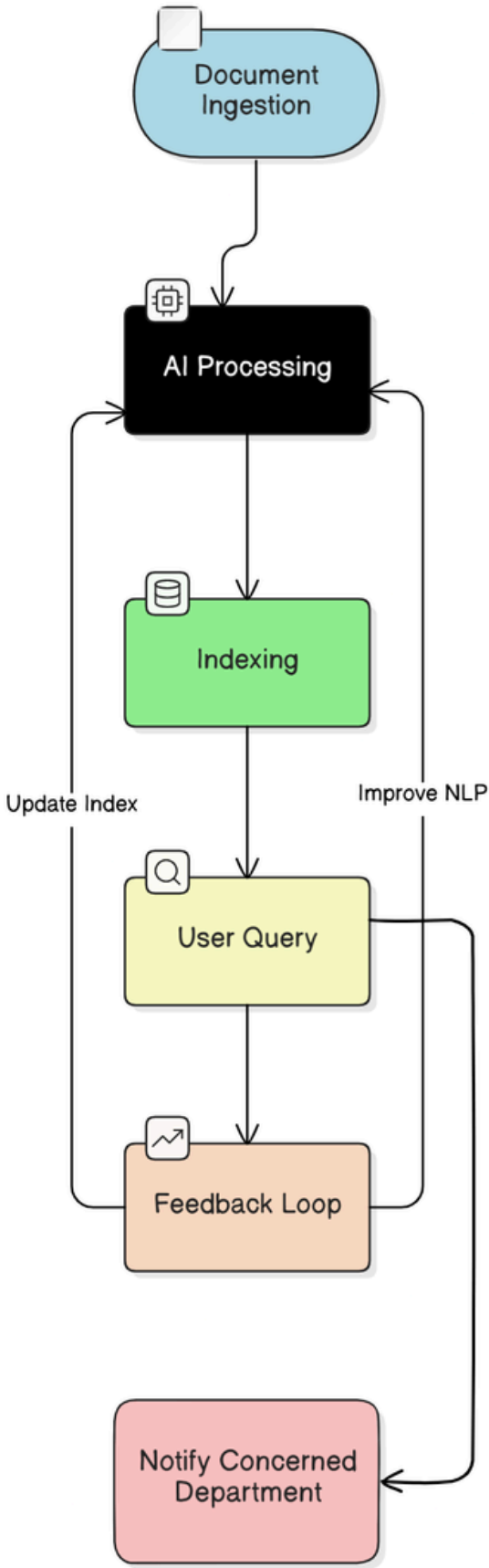
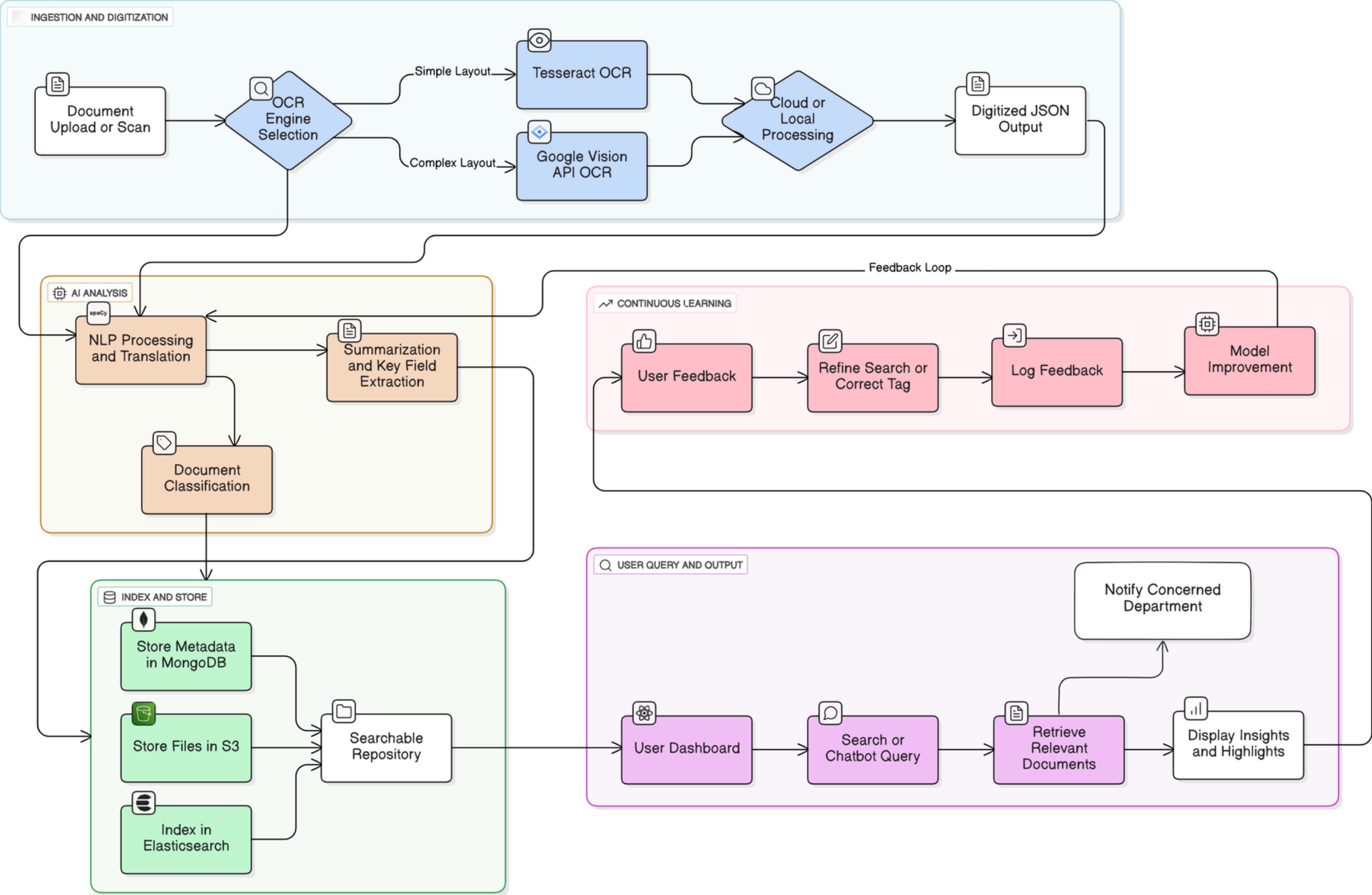


Alerts & Workflow Automation: Auto-detect critical items (deadlines, approvals) and notify relevant officers. Integrates with email/portal to remind about pending tasks, reducing missed actions.



On-the-Fly Translation: Input: Raw documents in any language → Translate to English (normalization) → Summarize via AI → Translate summary into user’s preferred language → Output: Multilingual dashboard & chatbot.

TECHNICAL APPROACH



FEASIBILITY AND VIABILITY



Feasibility:

- Proven Tech: OCR & NLP already reliable, just fine-tuned.
- Scalable Cloud: Easily handles growing document volumes.
- Seamless Integration: Works with DDFS/SharePoint via APIs.
- Low Cost: Open-source + APIs ensure quick, affordable setup.



Viability:

- Sustainability Viability: Runs long-term with low cost and upkeep.
- Scalability Viability: Expands easily across metros and workloads.
- Adoption Viability: User-friendly, quick to learn for staff.
- Impact Viability: Saves time, boosts compliance and efficiency.



Challenges:

- Volume Overload → Lakhs of diverse files may overwhelm system & AI.
- OCR Accuracy → Handwritten/old scans reduce recognition quality.
- User Adoption → Staff may resist shifting from manual filing.
- Data Security → Sensitive contracts & directives must be protected.



Solutions:

- Smart Indexing → Elasticsearch, archiving & batch processing.
- Hybrid OCR → Multi-engine OCR with KMRL training & flags.
- Phased Rollout → Simple UI, workshops & pilot adoption.
- Robust Security → Role-based access, encryption & audits.



Use Cases:

- Regulatory Compliance: Summarizes directives & circulars to meet deadlines.
- Financial Processing: Summarizes invoices & POs for faster payments.
- Maintenance Decisions: Summarizes reports & job cards for quick readiness.
- Knowledge Management: Converts policies & minutes into searchable summaries.



Business Potential:

- Scalable SaaS: License the platform to metros, PSUs, and govt. bodies.
- Cross-Sector Expansion: Deploy in sectors like healthcare, utilities, and banking.
- Analytics Service: Offer insights and trends from processed document data.
- Localization Edge: Multilingual support boosts state and central govt. adoption.

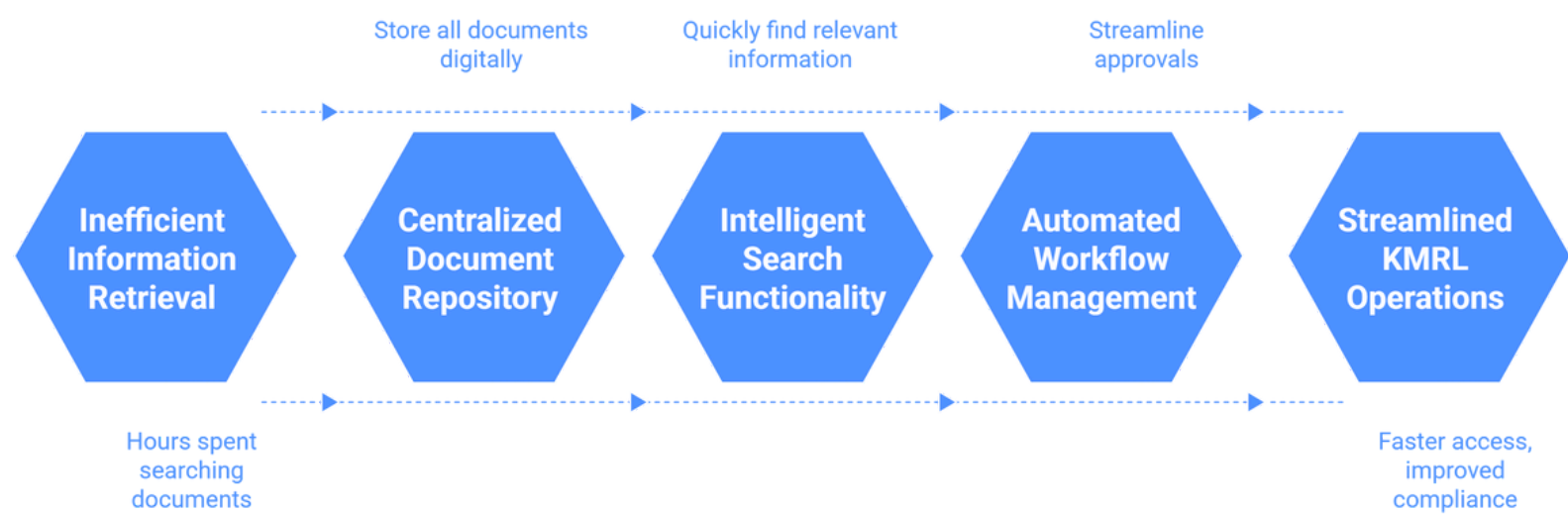
★ Supporting Facts for Feasibility and Viability ★

- A 2023 report by IDC found that employees spend up to 30% of their time searching for information, showing the urgent need for smarter document management.
- A study by Gartner highlighted that 80% of enterprise data is unstructured (PDFs, scans, emails), making AI-powered summarization and search critical.
- According to Deloitte, organizations lose nearly \$2.5 million annually due to compliance failures caused by missed regulatory updates.
- A survey by McKinsey found that effective knowledge management systems can improve productivity by 20–25%, directly boosting operational efficiency.

IMPACT AND BENEFITS

Potential Impact

- **Empowering Staff:** KochiDocs cuts search time by ~80%, saving clerks ~1.5 hrs/day. Staff can focus on planning and service instead of paperwork, boosting morale and productivity .
- **Data-Driven Decisions:** Centralized access enables 2–4× faster approvals and workflows, allowing quick responses to operational needs.
- **Compliance & Transparency:** Every document is logged, ensuring no lost files, audit readiness, and reduced regulatory risks.
- **Passenger Impact:** Faster decisions and better maintenance reduce train downtimes, improving service reliability and customer experience.

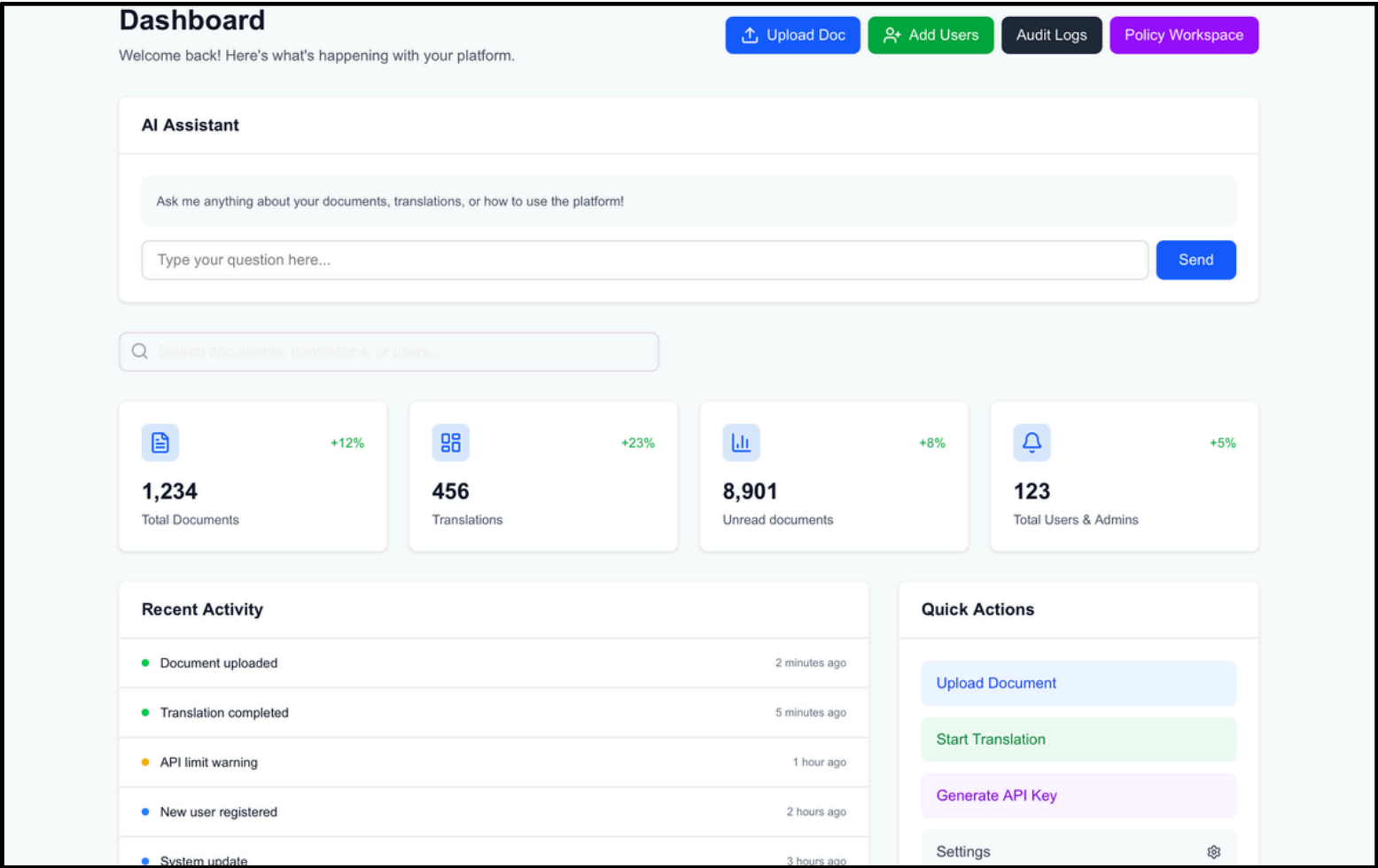


Benefits

- **Social Benefits**
 - More reliable passenger service through faster issue resolution.
 - Higher staff morale and productivity with less frustration.
 - Greater transparency builds public trust.
- **Economic Benefits**
 - Savings on paper, printing, and storage costs.
 - 2–4× faster workflows optimize staff use.
 - Quicker decisions improve service uptime and revenue.
 - Lower training costs as new staff can access well-organized records easily.
- **Environmental Benefits**
 - ~90% less paper and ink use; no of trees saved yearly.
 - Supports sustainability and green initiatives.
 - Digital archiving preserves knowledge without resource strain.
 - Reduced carbon footprint from fewer physical document movements.
- **Strategic Benefits**
 - Advances Digital India and Atmanirbhar Bharat.
 - Aligns with UN SDG 9 by modernizing infrastructure.

RESEARCH AND REFERENCES

- [1] [Global-engineering-and-consultancy-case-study.pdf](#)
- [2] [AI-Powered Document Intelligence Platform for Kochi Metro Rail](#)
- [3] [Various Survey Statistics: Workers Spend Too Much Time Searching for Information - Cottrill Research](#)
- [4] [Kochi Metro to go paperless thanks to Technopark firm](#)



Feature	<i>KochiDocs</i>	KMRL – DDFS (Ospyn)	Gov/Compliance AI
Multilingual	✓	✗	✗
Explainability / Trust	✓	✗	✓
AI Summarization for Transit Ops Docs	✓	✗	✓
Front-Line Friendly UX	✓	✗	✗
WhatsApp/Scanned PDF Handling	✓	✓	✗
Rule + AI Hybrid Routing	✓	✓	✗
Compliance Deadline Extraction & Alerts	✓	✗	✓
Semantic Q&A (Vector Search)	✓	✗	✗